INCH POUND
A-A-58035
October 17, 1994

*COMMERCIAL ITEM DESCRIPTION*

*GENERATOR,SIGNAL*
*2.0 TO 8 4 GHZ*

The General Services Administration has authorized
the use of this Commercial Item Description (CID) for all federal agencies

## 1. SCOPE

This CID describes a synthesized signal generator which operates from 2 0 GHz to 8 4 GHz and has AM, FM and pulse modulation capabilities This is a general purpose signal generator This CID is meant as a minimum requirement for the synthesized signal generator in which only those manufacturers that meet or surpass the following requirements are qualified per this CID

## 2. SALIENT CHARACTERISTICS

2 1 Design and Construction. The equipment shall comply with the type III safety and environmental requirements of MIL-T-28800 The equipment shall also meet the following paragraphs on MIL-T-28800

| Paragraph | Title |
|---|---|
| 3.6.3 | Enclosure form, Style C |
| 3.7 4 1 | Vibration, sinusoidal |
| 3.8.3 | EMI, Type III |
| 3 9.3 1 | Accessible Potentials Type III |
| 3 9 3 1 2 | Leakage Current Protection, Type III |

Beneficial comments, recommendations, additions, deletions clarifications etc., and any data which may improve this document should be sent to The Resources and Logistics Services Division, SA-ALC/TILDD, Bldg 171, Post C-12, 485 Quentin Roosevelt Rd , Kelly AFB TX 78241-6524

AMSC N/A FSC 6625

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2.2 Electrical Power Sources The equipment shall operate from nominal commercial, military and shipboard power sources of 100, 110, 120, 200, 220, 240 volts AC (±10%) at line frequencies of 50 Hz to 400 Hz Power consumption shall not exceed 300 watts Transient state conditions shall meet MIL-T-28800 requirements for the classification specified herein

2 3 Mean time between failures (MTBF). The equipment MTBF shall be 2,000 hours The lower test MTBF shall be 2,000 hours and the upper test MTBF shall be 4,000 hours

2 4 Pre-conditioning requirement. Each deliverable equipment shall be capable of being subjected to the environmental screening effects of burn in and thermal cycling with thermal cycling being accomplished in the sample testing phase

2 5 Calibration and maintenance adjustments. The design of the system shall provide for readily accessible calibration adjustments and maintenance adjustments These adjustments shall be provided by variable value components which are adjustable by the use of simple means The calibration by substitution of selected components or parts is unacceptable unless specifically approved The calibration adjustments, wherever possible, shall be accessible without removal of the instrument case or modules The calibration interval shall be a period of one year or greater based on an operating time of 2,000 hours

2 6 Performance characteristics. The performance characteristics as described below shall constitute minimum Air Force requirement of this item and shall be adhered to or surpassed in quality and function

(a) Frequency The minimum frequency range shall be 2 GHz to 8 4 GHz

(1) Frequency accuracy shall be the same as internal time base

(2) Internal time base stability snall be a minimum of $4 \times 10^{-7}$ per day

(3) External time base input snall override the internal time base and accept a 10 MHz frequency with a level or 0 5 V RMS Stability shall be better than $1 \times 10^{-7}$

(b) Spectral purity. The spectral purity of the signal generator shall meet or exceed the specifications as follows

(1) Single-sideband phase noise shall be less than -70 dBc in a 1 Hz bandwidth, 1 KHz offset from the carrier in the CW mode

(2) Harmonics shall be less than -40 dBc at the +8 dBm output

(3) Sub Harmonics shall be less than -35 dBc at the +8 dBm output

(4) Spurious signals which are non-harmonic or power line related shall be less than -55 dBc for all carrier frequencies and shall be measured 15 KHz away from the carrier.



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(5) Power line related signals related to power line frequencies or any other internal mechanical or electrical devices shall be less than -45 dBc for any signal within +20 KHz from the fundamental frequency

(6) Residual AM shall be less than -50 dBc in a 300 Hz to 3 KHz post detection bandwidth

(7) Residual FM shall be less than 25 Hz RMS in a 300 Hz to 3 KHz post-detection bandwidth

(c) Resolution. The resolution shall be 1 KHz to 6 6 GHz and 2 KHz to 8 4 GHz.

(d) Power output. The power output level shall be adjustable from +8 dBm to -120 dBm over the entire operating temperature range into a 50 ohm load with a standing wave ratio of 2.5 or less The output shall have a flatness of +2 dB at 3 dBm output The absolute level accuracy shall be a minimum of +4.5 dB The output shall have a resolution of 0 1 dB

(e) External modulation inputs. The signal generator shall be able to accept the following inputs

(1) Pulse modulation. The pulse input shall accept a modulation signal with frequencies from 50 Hz to 500 KHz with a 15 nanoseconds to 2 microsecond rise time and a pulse width of 25 ns to 10 µsec The on/off ratio shall be a minimum of 60 dBm

(2) AM. The external AM shall accept modulation rates of 10 Hz to 50 KHz with a depth of 75% The AM sensitivity shall be 1 v p-p into a 600 ohm load at 50% AM modulation and a 1 KHz rate

(3) FM. The external FM shall accept an FM rate from 10 Hz to 1 MHz and an FM deviation of at least 2 MHz The FM sensitivity shall be at least 1 volt peak into a 50 ohm input and the distortion shall be less than 5% for all FM rates and all deviations The distortion will be measured at 10 Hz, 500 KHz, and 1 MHz rates and 750 KHz and 1 8 MHz deviation

(f) Protection. The signal generator shall contain input/output internal power protection which will allow full functionality after being subjected to a 1 watt reverse power signal at the RF output for a period of 1 minute

(g) Signal connections. All input and output connections and controls shall be available on the front of the signal generator with the exception of GPIB and external time base

(h) Programmable. The signal generator shall be front panel or remotely programmable except power on/off The remote shall be IEEE 488 as specified in Section 3.3 4 of MIL-T-28800



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(i) Data transfer. All numeric and alphanumeric data (input and output) shall be American Standard Code for Information Interchange (ASCII) and the most significant digit shall be transferred first

(j) Operating temperature. Operating temperature shall be +10 to 45°C

(k) Manuals. The recommended support equipment list contained in the maintenance manual shall contain a generic description of the equipment with minimum parameter requirements for testing and calibrating the signal generator The listing shall not contain manufacturer's name or part number

## 3. REGULATORY REQUIREMENTS

In accordance with section 23 403 of the Federal Acquisition Regulations, the Government's policy is to acquire items composed of the highest percentage of recovered materials practicable, consistent with maintaining a satisfactory level of competition without adversely affecting performance requirements or exposing the supplier's employees to undue hazards from the recovered materials

## 4. QUALITY ASSURANCE PROVISIONS

Certification. The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this commercial item description and that the product conforms to the producer's own drawings, specifications, standards and quality assurance practices and is the same product offered for sale in the commercial marketplace The government reserves the right to require proof of such conformance prior to first delivery, and thereafter, as may be otherwise provided for under the provisions of the contract

The contracting officer has the option of accepting or rejecting the product

## 5. PACKAGING AND MARKING

The items shall be packed in accordance with normal commercial practice and packed to assure acceptance by common carrier and provide product protection against loss and damage during multiple shipments, handling, and storage The shipping shall be in compliance with national motor freight classification and uniform freight classification Shipping container shall be marked as specified in the contract or order

## 6. NOTES

6.1 Metric Products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within the specified tolerances using conversion tables contained in the latest revision of Federal Standard No 376, and all other requirements of this CID are met If a product is manufactured to metric dimensions exceeding the tolerances specified in the inch-pound units, a request should be made to the contracting officer to determine if the product is acceptable



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6 2 Federal and military specifications are available from the Defense Documents Order Desk, Building 4D, 700 Robins Avenue, Philadelphia, PA 19111-5094

6 3 Known Acceptable Products. The following product(s) are known to have previously met the salient characteristics of this CID and need not resubmit bid samples or certification All other manufacturers must meet the requirements of this CID and submit to bid testing and certification as required

| Manufacturer | Cagec | Model | NSN |
|---|---|---|---|
| Giga-tronics | 58900 | 605/2-8 4 (171) | 6625-01-302-2712 |

MILITARY INTERESTS

PREPARING ACTIVITY
Air Force - 82

Custodian.
Air Force - 99

Reviewing activities.
DLA-ES

Project No 6625-0711



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